

Finite-dimensional domains in the completeness problem for $CL(X, Y)$

Partial packet for arXiv:1612.01364

Source and target

The source is Antonin Dalet and Gilles Lancien, *Some properties of coarse Lipschitz maps between Banach spaces*, arXiv:1612.01364 [1]. In Section 4 the source introduces the net extension property (NEP), proves in Proposition 4.4 that NEP implies the Lipschitz representation property (LRP), and proves in Proposition 4.5 that LRP implies completeness of the quotient normed space $(CL(X, Y), \|\cdot\|_{CL})$. The section then leaves open whether this space is always complete.

Proposition 4.5. *Assume that X and Y are Banach spaces such that (X, Y) has the μ -LRP for some $\mu \geq 1$. Then $(CL(X, Y), \|\cdot\|_{CL})$ is a Banach space*

Proof. Let $(f_n)_{n=1}^\infty$ be a sequence in $\mathcal{CL}(X, Y)$ such that $\sum_{n=1}^\infty \|f_n\|_{CL} < \infty$. Then for any n in $\mathbb{N}$, there exists $g_n \in Lip_0(X, Y)$ such that g_n belongs to the equivalence class of f_n and $Lip(g_n) \leq \mu\|f_n\|_{CL} + 2^{-n}$. Then using the completeness of $(Lip_0(X, Y), \|\cdot\|_L)$ we get that there exists $g \in Lip_0(X, Y)$ such that

$$\lim_{N \rightarrow \infty} \|g - \sum_{n=1}^N g_n\|_L = 0.$$

It follows that $\lim_{N \rightarrow \infty} \|g - \sum_{n=1}^N f_n\|_{CL} = 0$, which concludes our proof. $\square$

Remark. We do not know if $(CL(X, Y), \|\cdot\|_{CL})$ is a Banach space without any assumption on the Banach spaces X and Y . We conjecture that it is not the case, but a counterexample still has to be constructed.

The target in this packet is the finite-dimensional-domain subcase of the open completeness question.

Result

Theorem 1. *Let X be a finite-dimensional Banach space and let Y be any Banach space. Then (X, Y) has the net extension property of Definition 4.1 in [1]. Consequently, by Propositions 4.4 and 4.5 of [1], $(CL(X, Y), \|\cdot\|_{CL})$ is a Banach space.*

Remark 1. *Thus any counterexample to the source's general conjecture must have infinite-dimensional domain. The argument does not address the infinite-dimensional case. The source's separate question whether LRP/ANEP implies NEP is handled in a companion full packet.*

Proof

Fix a finite-dimensional Banach space X . Let $\mathcal{M} \subset X$ be a maximal 1-separated set. Then $\mathcal{M}$ is a 1-net in X . Since X is finite-dimensional, $\mathcal{M}$ has bounded local multiplicity: for every $R > 0$ there is an integer N_R such that

$$\#(\mathcal{M} \cap B(x, R)) \leq N_R \quad (x \in X).$$

Indeed, the balls $B(m, 1/2)$, $m \in \mathcal{M} \cap B(x, R)$, are disjoint and are contained in $B(x, R + 1/2)$, so comparison with Lebesgue measure on X gives a bound independent of x .

It remains to show that every Lipschitz map $h : \mathcal{M} \rightarrow Y$ admits a Lipschitz extension to X . Write $L = \text{Lip}(h)$. For $m \in \mathcal{M}$ define

$$\eta_m(x) = (2 - \|x - m\|)_+, \quad S(x) = \sum_{m \in \mathcal{M}} \eta_m(x), \quad \lambda_m(x) = \frac{\eta_m(x)}{S(x)}.$$

The sums are finite because of local multiplicity. Since $\mathcal{M}$ is a 1-net, $S(x) \geq 1$ for all x . Also $S(x) \leq 2N_2$. The functions λ_m form a Lipschitz partition of unity, supported in $B(m, 2)$. More precisely, there is a constant $A_X < \infty$, depending only on the packing numbers of $\mathcal{M}$, such that for all $x, y \in X$,

$$\sum_{m \in \mathcal{M}} |\lambda_m(x) - \lambda_m(y)| \leq A_X \|x - y\| \quad \text{whenever } \|x - y\| \leq 1. \quad (1)$$

To see this, note that for $\|x - y\| \leq 1$ all indices appearing in either partition lie in $B(x, 3)$, hence there are at most N_3 of them. The maps η_m are 1-Lipschitz, and S is Lipschitz on such pairs with constant controlled by N_3 , while $S \geq 1$.

Define the partition extension

$$H_0(x) = \sum_{m \in \mathcal{M}} \lambda_m(x) h(m).$$

We first check that H_0 is Lipschitz. Suppose $\|x - y\| \leq 1$, and choose $p \in \mathcal{M}$ with $\|x - p\| \leq 1$. Since $\sum_m (\lambda_m(x) - \lambda_m(y)) = 0$, we may subtract $h(p)$ and get

$$H_0(x) - H_0(y) = \sum_m (\lambda_m(x) - \lambda_m(y)) (h(m) - h(p)).$$

For every index in this finite sum, m is within distance 4 of p . Thus, using (1),

$$\|H_0(x) - H_0(y)\| \leq 4A_X L \|x - y\|.$$

For $\|x - y\| > 1$, choose $p, q \in \mathcal{M}$ with $\|x - p\| \leq 1$ and $\|y - q\| \leq 1$. The support condition gives $\|H_0(x) - h(p)\| \leq 3L$ and $\|H_0(y) - h(q)\| \leq 3L$, while $\|h(p) - h(q)\| \leq L(\|x - y\| + 2)$. Hence

$$\|H_0(x) - H_0(y)\| \leq L(\|x - y\| + 8) \leq 9L\|x - y\|.$$

So H_0 is Lipschitz, with constant bounded by a constant depending only on X times L .

The map H_0 need not agree with h on $\mathcal{M}$, but the error is uniformly bounded. If $m \in \mathcal{M}$, then $H_0(m)$ is a convex combination of values $h(n)$ with $\|n - m\| < 2$. Hence

$$e(m) := h(m) - H_0(m) \quad \text{satisfies} \quad \|e(m)\| \leq 2L.$$

Correct this error with disjoint bumps. Put

$$\beta_m(x) = (1 - 2\|x - m\|)_+, \quad E(x) = \sum_{m \in \mathcal{M}} \beta_m(x) e(m).$$

The supports of the positive parts of the functions β_m are disjoint, since $\mathcal{M}$ is 1-separated. On each support E is $4L$ -Lipschitz. If two points lie in different supports, the line segment between

them leaves the first support before entering the second, and the corresponding bump value is zero at the exit and entry points; the same $4L$ bound follows by applying the one-support estimate to the two subsegments. Thus E is globally Lipschitz.

Finally set $H = H_0 + E$. Then $H : X \rightarrow Y$ is Lipschitz and, for every $m \in \mathcal{M}$,

$$H(m) = H_0(m) + e(m) = h(m),$$

because $\beta_m(m) = 1$ and $\beta_n(m) = 0$ for $n \neq m$. Therefore $(\mathcal{M}, X, Y)$ has the Lipschitz extension property. Since $\mathcal{M}$ is a net in X , the pair (X, Y) has NEP. Proposition 4.4 of [1] gives LRP, and Proposition 4.5 then gives completeness of $(CL(X, Y), \|\cdot\|_{CL})$.

Verification and limitations

The exact arXiv id and the core phrases `coarse Lipschitz`, `CL(X,Y) completeness`, `net extension property`, and `LRP` were checked in the run indexes before packaging. No previous packet for arXiv:1612.01364 was found. Nearby coarse-Lipschitz entries in the run concern different permanence or renorming questions and do not cover this finite-dimensional NEP argument.

A related prior run artifact, `1506.05391_finite_dimensional_net_extensions`, records the same finite-dimensional net-extension mechanism for Naor’s Rosendal-net question, using later literature. Thus this packet should not be read as claiming novelty of the finite-dimensional extension lemma itself. The target-level contribution here is the application of NEP to Dalet–Lancien’s completeness problem through their Propositions 4.4 and 4.5.

This packet is a partial result only. It does not construct the conjectured counterexample and it does not settle completeness for infinite-dimensional domains. The adjacent LRP/ANEP-versus-NEP converse question is treated in the companion full packet for arXiv:1612.01364.

Human-review recommendation

Likely valid partial result. Review should check the partition-of-unity extension estimate, especially the disjoint-bump correction step. The deduction from NEP to completeness is exactly the source’s Proposition 4.4 followed by Proposition 4.5.

References

- [1] A. Dalet and G. Lancien, *Some properties of coarse Lipschitz maps between Banach spaces*, arXiv:1612.01364.